

# Target validation paths — demo-nscclc-egfr-l858r-post-osu-d3m0

*The diagnostic and biomarker workup that hardens the targetable-feature call*

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# Target validation paths

EGFR L858R, MET amplification (GCN 8.2), and T790M-absent are all confirmed at decision-relevant resolution, so no diagnostic test gates the savolitinib + osimertinib trial enrollment (SAFFRON / SACHI). The validation work here is about resistance characterization at progression, not target confirmation. The goal is to make sure the case really is MET-amplified bypass resistance and not something more complicated wearing the same imaging signature.

## EGFR L858R

The most consequential miss right now would be SCLC histologic transformation, which is the highest-PPV resistance pattern that no liquid biopsy can confirm. Tumor NGS including TP53 and RB1 status is the screen: co-loss is the strongest predictor of transformation, with reported rates of 3–14% across post-osimertinib cohorts. If imaging morphology shifts or co-loss is detected, fresh re-biopsy with a neuroendocrine IHC panel (chromogranin, synaptophysin, INSM1, Ki-67) is the only way to confirm. ctDNA covering EGFR C797S and exon-20 inserts is the parallel on-target resistance check; a positive C797S result reframes the next-line conversation toward fourth-generation EGFR-TKIs or combination strategies.

## MET amplification

The FISH GCN 8.2 result clears SAFFRON's enrollment threshold ( $\geq 6$ ) cleanly, but MET amplification can be focal, and primary-vs-metastatic discordance is a documented confounder in this setting. A second-site FISH (or a matched archival block) refines confidence that the dominant disease, not just the biopsied site, is the amplified one. MET IHC (clone SP44) is the orthogonal-modality complement to the SAFFRON eligibility profile. A comprehensive NGS panel that picks up HER2 / BRAF / FGFR co-alterations frames durability expectations and the next-line plan.

## Where to order these assays

The preferred provider for each assay is marked (**preferred**), selected on company size, reputation, US-based location, and turnaround time. Other providers in the row are listed in case the preferred lab is unreachable for this patient.

| Assay                                      | Provider                                          | Decision gated                                                                                                                 | Contact                                                                  |
|--------------------------------------------|---------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| ctDNA panel (EGFR C797S / T790M / exon-20) | Guardant Health (preferred)<br>(Guardant360 CDx)  | Distinguishes on-target (C797S, exon-20) vs bypass post-osimertinib resistance; informs fourth-gen TKI vs combination strategy | test info · 505 Penobscot Drive, Redwood City, CA 94063 · 1-855-698-8887 |
| ctDNA panel                                | Foundation Medicine<br>(FoundationOne Liquid CDx) | Distinguishes on-target vs bypass post-osimertinib resistance                                                                  | test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639      |
| ctDNA panel                                | Tempus (xF+)                                      | Distinguishes on-target vs bypass post-osimertinib resistance                                                                  | test info · 600 West Chicago Avenue, Chicago, IL 60654 · 1-800-739-4137  |

|                                                      |                                                            |                                                                                                                            |                                                                                |
|------------------------------------------------------|------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| ctDNA panel                                          | Caris Life Sciences (Caris Assure)                         | Distinguishes on-target vs bypass post-osimertinib resistance                                                              | test info · 4610 South 44th Place, Phoenix, AZ 85040 · 1-888-979-8669          |
| ctDNA panel                                          | NeoGenomics Laboratories (NeoLAB Liquid)                   | Distinguishes on-target vs bypass post-osimertinib resistance                                                              | test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907        |
| <b>Tumor NGS for TP53 / RB1 + bypass alterations</b> | <b>Foundation Medicine (preferred) (FoundationOne CDx)</b> | <b>Predicts SCLC histologic transformation; informs whether next-line stays adenocarcinoma-targeted or pivots to chemo</b> | <b>test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639</b>     |
| Tumor NGS                                            | Caris Life Sciences (Molecular Intelligence)               | Predicts SCLC histologic transformation                                                                                    | test info · 4610 South 44th Place, Phoenix, AZ 85040 · 1-888-979-8669          |
| Tumor NGS                                            | Tempus (xT)                                                | Predicts SCLC histologic transformation                                                                                    | test info · 600 West Chicago Avenue, Chicago, IL 60654 · 1-800-739-4137        |
| Tumor NGS                                            | NeoGenomics Laboratories (NeoTYPE Comprehensive)           | Predicts SCLC histologic transformation                                                                                    | test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907        |
| Tumor NGS                                            | Memorial Sloan Kettering (MSK-IMPACT)                      | Predicts SCLC histologic transformation                                                                                    | test info · 1275 York Avenue, New York, NY 10065 · 1-212-639-2000              |
| <b>Re-biopsy + neuroendocrine IHC panel</b>          | <b>Mayo Clinic Laboratories (preferred)</b>                | <b>Confirms SCLC transformation when imaging shifts or TP53/RB1 co-loss is detected; pivots therapy entirely</b>           | <b>test info · 200 First Street SW, Rochester, MN 55905 · 1-800-533-1710</b>   |
| Re-biopsy + neuroendocrine IHC                       | ARUP Laboratories                                          | Confirms SCLC transformation when imaging shifts or TP53/RB1 co-loss is detected                                           | test info · 500 Chipeta Way, Salt Lake City, UT 84108 · 1-800-242-2787         |
| Re-biopsy + neuroendocrine IHC                       | LabCorp / Esoterix Oncology                                | Confirms SCLC transformation when imaging shifts or TP53/RB1 co-loss is detected                                           | test info · 358 South Main Street, Burlington, NC 27215 · 1-800-345-4363       |
| Re-biopsy + neuroendocrine IHC                       | Quest Diagnostics                                          | Confirms SCLC transformation when imaging shifts or TP53/RB1 co-loss is detected                                           | test info · 500 Plaza Drive, Secaucus, NJ 07094 · 1-866-697-8378               |
| Re-biopsy + neuroendocrine IHC                       | Memorial Sloan Kettering Diagnostic Molecular Pathology    | Confirms SCLC transformation when imaging shifts or TP53/RB1 co-loss is detected                                           | test info · 1275 York Avenue, New York, NY 10065 · 1-212-639-2000              |
| <b>MET FISH (gene/CEP7 ratio)</b>                    | <b>NeoGenomics Laboratories (preferred)</b>                | <b>Refines confidence in MET amp gating result for SAFFRON enrollment; does not gate eligibility</b>                       | <b>test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907</b> |
| MET FISH                                             | Foundation Medicine                                        | Refines MET amp gating result; does not gate eligibility                                                                   | test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639            |

|                                                             |                                                            |                                                                                                                     |                                                                              |
|-------------------------------------------------------------|------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
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| MET FISH                                                    | LabCorp / Esoterix Oncology                                | Refines MET amp gating result; does not gate eligibility                                                            | test info · 358 South Main Street, Burlington, NC 27215 · 1-800-345-4363     |
| MET FISH                                                    | Quest Diagnostics                                          | Refines MET amp gating result; does not gate eligibility                                                            | test info · 500 Plaza Drive, Secaucus, NJ 07094 · 1-866-697-8378             |
| <b>MET IHC (clone SP44)</b>                                 | <b>Mayo Clinic Laboratories (preferred)</b>                | <b>Confirms MET protein-level expression; orthogonal complement to FISH but does not change SAFFRON eligibility</b> | <b>test info · 200 First Street SW, Rochester, MN 55905 · 1-800-533-1710</b> |
| MET IHC                                                     | ARUP Laboratories                                          | Confirms MET protein-level expression                                                                               | test info · 500 Chipeta Way, Salt Lake City, UT 84108 · 1-800-242-2787       |
| MET IHC                                                     | NeoGenomics Laboratories                                   | Confirms MET protein-level expression                                                                               | test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907      |
| MET IHC                                                     | LabCorp / Esoterix Oncology                                | Confirms MET protein-level expression                                                                               | test info · 358 South Main Street, Burlington, NC 27215 · 1-800-345-4363     |
| MET IHC                                                     | Quest Diagnostics                                          | Confirms MET protein-level expression                                                                               | test info · 500 Plaza Drive, Secaucus, NJ 07094 · 1-866-697-8378             |
| <b>Comprehensive bypass NGS (HER2 / BRAF / KRAS / FGFR)</b> | <b>Foundation Medicine (preferred) (FoundationOne CDx)</b> | <b>Frames durability expectations and the next-line plan for MET-directed therapy; does not gate SAFFRON</b>        | <b>test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639</b>   |
| Comprehensive bypass NGS                                    | Caris Life Sciences (Molecular Intelligence)               | Frames durability and next-line plan for MET-directed therapy                                                       | test info · 4610 South 44th Place, Phoenix, AZ 85040 · 1-888-979-8669        |
| Comprehensive bypass NGS                                    | Tempus (xT)                                                | Frames durability and next-line plan for MET-directed therapy                                                       | test info · 600 West Chicago Avenue, Chicago, IL 60654 · 1-800-739-4137      |
| Comprehensive bypass NGS                                    | NeoGenomics Laboratories (NeoTYPE Comprehensive)           | Frames durability and next-line plan for MET-directed therapy                                                       | test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907      |
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*Decision support, not medical advice. Confirm assay availability and current standards with the treating team and the local pathology service.*